Cardiac Contractility Analysis

Jupyter Notebook — User Documentation

1. Overview

This Jupyter notebook analyses time-lapse fluorescence microscopy images of beating cardiac tissue. It extracts quantitative measures of contractile motion — specifically contraction amplitude and contraction speed — as traces over time, and saves results as numerical data files, plots, and image stacks.

The notebook is divided into two cells:

- Cell 1 — Parameter Setup: a graphical form where you enter all analysis settings before running
- Cell 2 — Analysis Pipeline: the processing code that runs automatically once parameters are submitted

Both single-file and batch processing modes are supported. When batch mode is enabled, the pipeline processes every compatible image file in the specified folder in sequence.

2. Requirements

2.1 Software

- Python 3.11 (3.9 or later)
- JupyterLab or Jupyter Notebook
- The following Python packages must be installed in your environment:
  - napari — image viewer for interactive display in single-file processing mode
  - numpy — numerical array operations
  - tifffile — reading and writing TIFF/OME-TIFF files
  - matplotlib — plotting
  - scipy — signal processing (FFT, Savitzky-Golay filter, peak detection)
  - ipywidgets — interactive form widgets in the notebook
  - bioio, bioio-czi, bioio-bioformats — reading CZI and other microscopy file formats

> **Tip:** *Install all packages into a dedicated conda or virtual environment before launching the notebook. Running pip install bioio bioio-czi bioio-bioformats napari tifffile scipy matplotlib ipywidgets in your environment terminal will install the core dependencies. See here for more detailed instructions.*

2.2 Input Files

- One or more time-lapse fluorescence images of beating cardiac tissue

- Supported formats: .czi (Zeiss) and .ome.tiff / .tif / .tiff, anything BioFormats can open (though not tested)
- Images must be grayscale (single channel) or must have the relevant channel accessible as the first channel in the file
- Files must be placed in a single folder; sub-folders are not scanned

3. Cell 1 — Parameter Setup

Run Cell 1 to display the parameter form. Fill in all fields and click Submit parameters. The notebook will confirm that parameters have been accepted before you proceed to Cell 2.

> **Important:** *You must click Submit parameters before running Cell 2. If you change any value in the form, click Submit parameters again to update the settings.*

3.1 Parameter Reference

Parameter	Default	Description
Input directory	`Hardcoded path`	Full path to the folder containing your image file(s). Edit the text field to point to your data folder.
Process all files in directory	`Unticked`	When ticked, every file in the directory is processed in sequence. When unticked, only the file named in the File name field is processed.
File name	`Example.czi`	Required only when batch mode is not ticked. Enter the exact filename including extension (e.g. sample_01.czi).
Video rate (fps)	`1`	The frame rate at which the time-lapse was recorded, in frames per second. Check your acquisition metadata if unsure.
Calculate beat rate	`Ticked`	When ticked, the beat rate is estimated automatically from the image data using autocorrelation. When unticked, the value entered in Manual beat rate is used.
Manual beat rate (Hz)	`1`	Used only when Calculate beat rate is unticked. Enter the expected beat frequency of the tissue in Hz (beats per second).
Offset (frames)	`15`	The rolling window length used to compute pixel variance. Used only when Calculate offset is unticked.
Calculate offset	`Ticked`	When ticked, the offset is calculated automatically as: Fractional offset × (video rate ÷ beat rate).

Parameter	Default	Description
Fractional offset	`0.2`	Fraction of one beat period used to set the rolling window length when automatic offset calculation is enabled. A value of 0.2 means the window spans 20% of one beat cycle.
Calibrate with pixel size	`Unticked`	When ticked, speed and contraction outputs are multiplied by the pixel calibration value and expressed in physical units (µm and µm/s). When unticked, outputs remain in pixel units.
Pixel calibration (µm/px)	`1`	Physical size of one pixel in micrometres. Check your microscope acquisition settings or image metadata.

3.2 Choosing the Offset

The offset parameter controls the temporal window over which pixel-level variability is measured. It should be set to a fraction of one beat period — long enough to capture meaningful motion but short enough not to span multiple beats.

With automatic offset calculation enabled and the default fractional offset of 0.2, the window is set to 20% of the estimated beat period. For a tissue beating at 1 Hz recorded at 31 fps, this gives an offset of approximately 6 frames.

> **Tip:** *If the contraction trace looks noisy or flat, try adjusting the fractional offset. Values between 0.15 and 0.35 are typical. A larger fraction smooths the trace but may reduce sensitivity to rapid contractions.*

4. Cell 2 — Analysis Pipeline

Run Cell 2 after submitting parameters in Cell 1. The pipeline runs automatically for each file and prints progress messages to the notebook output. Do not modify the code in this cell.

4.1 Pipeline Steps

#	Step	Description
1	**Load image**	Opens the input file using tifffile (for .tif/.tiff) or bioio (for .czi and other formats). The image is transposed to Height × Width × Time format for processing.
2	**Beat rate estimation**	If Calculate beat rate is enabled: computes the mean absolute frame-to-frame pixel difference to produce a motion signal, detrends it, and applies autocorrelation to find the dominant beat period. The result is printed and saved to the parameters file. An autocorrelation plot is also saved.

#	Step	Description
3	**Pixel variance stack**	For each frame in the time series, computes the standard deviation of pixel intensities across a rolling window of length offset+1 frames. This produces a 3D stack (H × W × T) where high values indicate pixels that are moving rapidly. The stack is normalised to 16-bit range and saved as an OME-TIFF.
4	**Speed calculation**	Averages the pixel variance map at each time point across the whole image to produce a 1D speed trace. A baseline correction using a 10%-width uniform filter is applied to remove slow drift. The trace is optionally scaled by the pixel calibration.
5	**Key frame selection**	Identifies the frame with the lowest speed value as the reference quiet frame (key frame). This frame represents the tissue at rest and is used as the baseline for contraction calculation.
6	**Contraction calculation**	For each frame, computes the pixel-wise standard deviation within the rolling window and subtracts the equivalent statistic at the key frame. The mean absolute difference across all pixels gives the contraction value at that time point. A Savitzky-Golay filter (window ≈ one beat period, polynomial order 3) is applied to smooth the trace.
7	**Save results**	Speed and contraction traces are saved as tab-delimited .txt files containing time, raw values, scaled values (if calibration is enabled), and normalised values. Four PNG plots are also saved: scaled and normalised versions of both traces.
8	**Save PV stack**	The pixel variance stack is padded with zero frames at the start (to match the original time axis) and saved as an OME-TIFF with spatial and temporal metadata for opening in napari or Fiji.
9	**napari viewer**	In single-file mode, the original image stack and the pixel variance stack are opened in napari for interactive inspection. In batch mode, napari is not launched to avoid blocking the pipeline.
10	**Memory cleanup**	Large arrays are explicitly deleted after each file to free memory before the next file is processed.

5. Output Files

All outputs are saved to an analysis/ subfolder created automatically inside your input directory. Files are named after the input image filename (without extension).

Output File	Contents
`*_processing_params.txt`	Plain text record of all parameters used for the analysis, including the calculated beat rate if auto-detection was enabled.
`*_autocorrelation_for_beatrate.png`	Plot showing the motion signal over time and the autocorrelation function used to estimate the beat rate. The selected period is marked with a red dot; other candidate peaks are shown in orange. Saved only when automatic beat rate calculation is enabled.

Output File	Contents
`*_contraction_scaled.png`	Plot of the contraction trace over time in pixel or µm units (depending on calibration setting).
`*_contraction_normalized.png`	Plot of the contraction trace normalised to the range [0, 1].
`*_speed_scaled.png`	Plot of the speed trace over time in pixel/s or µm/s units.
`*_speed_normalized.png`	Plot of the speed trace normalised to the range [0, 1].
`*_contraction.txt`	Tab-delimited text file with columns: time (s), contraction in pixels, contraction in µm (if calibrated), contraction normalised.
`*_speed.txt`	Tab-delimited text file with columns: time (s), speed in pixels/s, speed in µm/s (if calibrated), speed normalised.
`*_PixelVariance.ome.tiff`	OME-TIFF image stack of the pixel variance at each time point, with physical pixel size and time increment encoded in the metadata. Can be opened in Fiji or napari.

6. Beat Rate Estimation

When automatic beat rate calculation is enabled, the pipeline estimates the beat frequency directly from the image data without requiring user input. The method works as follows:

1. The mean absolute frame-to-frame pixel difference is computed for each consecutive pair of frames to produce a motion signal over time.
2. A 10%-width uniform filter is subtracted from the motion signal to remove slow baseline drift while preserving the beat periodicity.
3. A Savitzky-Golay filter smooths the signal.
4. The autocorrelation of the smoothed, zero-mean signal is computed. The first prominent peak in the autocorrelation (beyond a minimum lag corresponding to 0.3 seconds) gives the beat period.
5. The beat rate is calculated as 1 / period and printed in both Hz and BPM.

Important: *Inspect the autocorrelation plot saved to the output folder to verify that the correct peak was selected. The red dot marks the detected period. If the wrong peak is selected, switch to manual beat rate entry and provide the correct value.*

7. Interpreting the Outputs

7.1 Contraction trace

The contraction trace represents how much the tissue has moved relative to its quietest state at each point in time. Peaks in the trace correspond to moments of maximum contraction. The baseline (between beats) should ideally be close to zero; a rising or oscillating baseline indicates wave propagation across the field of view or other spatial heterogeneity in the beating pattern.

Key features to inspect:

- Peak regularity — consistent peak heights indicate stable, rhythmic contraction
- Peak shape — a sharp upstroke and slower relaxation are characteristic of healthy cardiac tissue
- Baseline stability — a flat baseline between beats indicates the tissue is truly at rest between contractions

7.2 Speed trace

The speed trace measures how rapidly pixels are changing intensity across the whole field at each time point. It reflects the velocity of the contraction wave passing through the field of view rather than the absolute displacement. Peaks in the speed trace should precede or coincide with peaks in the contraction trace.

7.3 Pixel variance stack

Open the _PixelVariance.ome.tiff file in Fiji (File > Open) or napari to inspect the spatial distribution of motion at each time point. Bright regions in the stack correspond to pixels that are changing rapidly. This allows you to confirm that the correct tissue region is being captured and to identify any spatial heterogeneity in the contraction pattern.

8. Troubleshooting

The notebook fails to import modules

One or more required packages are not installed in the active Python environment. Activate the correct conda environment before launching Jupyter, and install any missing packages using pip install or conda install.

The CZI file cannot be opened

The pipeline first tries bioio-czi and falls back to bioio-bioformats if that fails. If both fail, ensure that both bioio-czi and bioio-bioformats are installed and that the file is not open in another application. Some very large CZI files may time out; try converting to OME-TIFF first using Fiji (File > Save As > OME-TIFF).

The contraction trace is flat or very noisy

This is usually caused by a mismatch between the offset window and the actual beat period. Try enabling automatic offset calculation, or manually set the offset to approximately 15–25% of the number of frames per beat. Also verify that the correct channel is being loaded — if the image has multiple channels, only the first is used.

The beat rate is detected incorrectly

Inspect the autocorrelation plot. If the red dot is on a secondary peak rather than the primary peak, the autocorrelation has picked up a harmonic of the true beat rate. Switch to manual beat rate entry and input the correct value from your experimental records.

Memory errors during batch processing

Large image files can exhaust available RAM, especially during pixel variance calculation. Reduce the number of files processed in a single batch run, close napari and other applications before running, or increase the available memory on the machine. The pipeline explicitly clears variables between files, but very large individual files may still cause issues.

No output files are produced for a file

Check the notebook output for error messages printed during processing of that file. Common causes include: the file not being found at the specified path, an unsupported file format, or an image with only one frame (which cannot be analysed as a time series).

9. Glossary

Term	Definition
Autocorrelation	A mathematical technique that measures how similar a signal is to a time-shifted version of itself. Used here to detect the repeating period of the beating signal.
Beat rate	The frequency at which the cardiac tissue contracts, expressed in Hz (beats per second) or BPM (beats per minute).
Contraction	The measured displacement or change in tissue structure relative to a quiet reference frame, used as a proxy for the contractile amplitude of the tissue.
Key frame	The time point identified as the moment of minimum motion (tissue most at rest), used as the reference for contraction calculation.
Normalised	Rescaled to the range [0, 1] by subtracting the minimum and dividing by the range, allowing comparison between recordings regardless of absolute signal magnitude.
Offset	The number of frames used as the rolling temporal window for pixel variance calculation. Controls the time scale over which local motion is measured.
OME-TIFF	Open Microscopy Environment TIFF. A standard image file format for microscopy data that encodes spatial and temporal metadata (pixel size, time interval) alongside the image.
Pixel calibration	The physical size of one image pixel, expressed in micrometres per pixel (µm/px). Used to convert pixel-based measurements into physical units.
Pixel variance	The standard deviation of pixel intensity values within a rolling time window. High pixel variance at a given location indicates that the pixel is fluctuating rapidly, reflecting local motion.
Savitzky-Golay filter	A smoothing filter that fits a polynomial to successive overlapping windows of the signal. Preserves the shape and height of peaks while reducing noise, making it well suited to asymmetric biological waveforms.

Term	Definition
Speed	A whole-image measure of motion velocity, derived by averaging the pixel variance map across all pixels at each time point.
Time series	A sequence of images acquired at regular time intervals, forming a 3D dataset (Height × Width × Time).